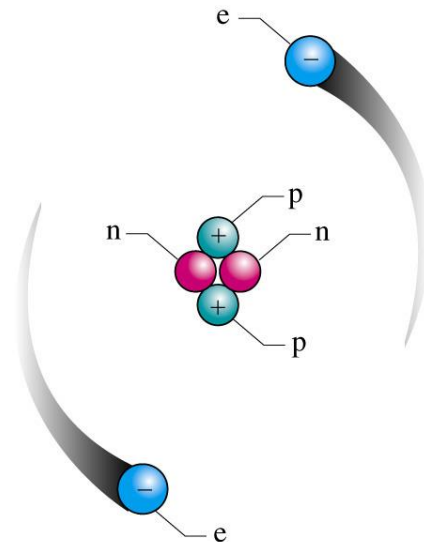


GENERAL CHEMISTRY I

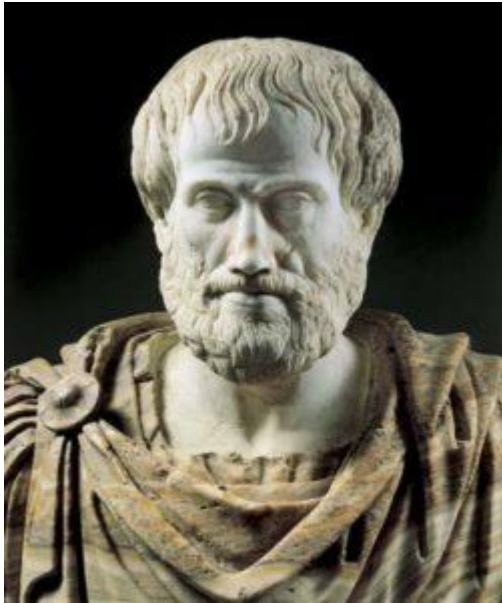


Chapter 2 Atoms, Molecules, and Ions

Contents

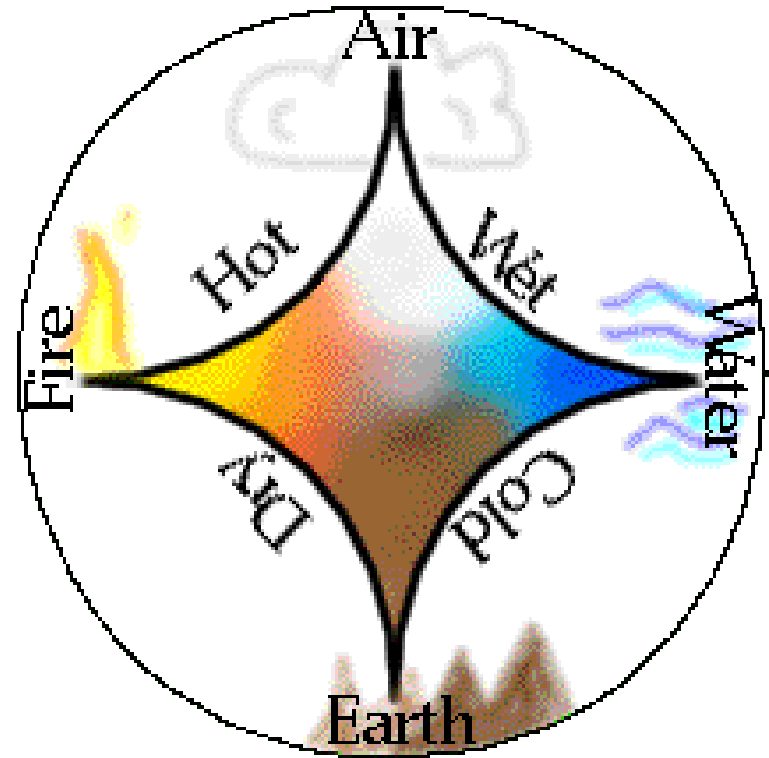
- 2-1 The Atomic Theory of Matter
- 2-2 The Discovery of Atomic Structure
- 2-3 The Modern View of Atomic Structure
- 2-4 Atomic Weights
- 2-5 The Periodic Table
- 2-6 Molecules and Molecular Compounds
- 2-7 Ions and Ionic Compounds
- 2-8 Naming Inorganic Compounds
- Self reading: 2-9 Some simple Organic Compounds

2-1 The Atomic Theory of Matter



Aristotle

(384 – 322 B.C.)

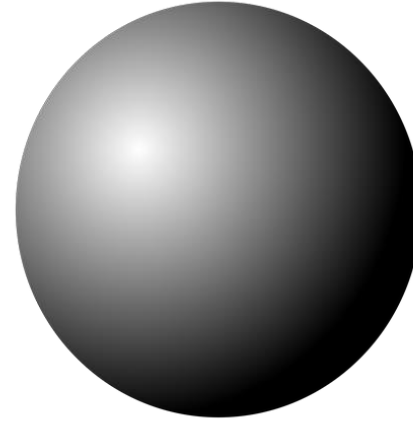


Four primary qualities of matter: hotness, coldness, wetness, and dryness

Their combinations appear as: **fire** (hot and dry), **earth** (dry and cold), **water** (cold and wet) and **air** (wet and hot)

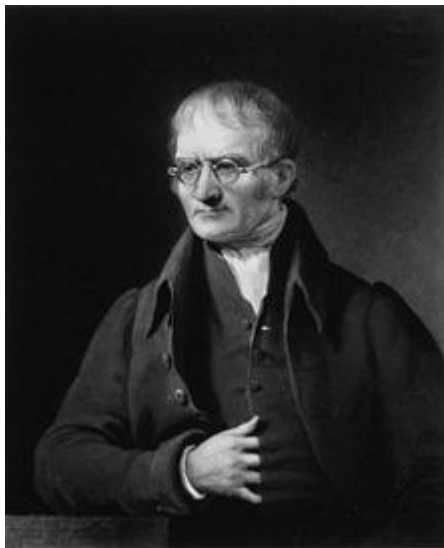


Democritus
(460 – 370 B.C.)



Atomos

- physically, but not geometrically, indivisible;
- indestructible;
- always in motion;
- there lies empty space between atoms;
- there is an infinite number of atoms and of kinds of atoms, which differ in shape and size;
- the more any indivisible exceeds, the heavier it is



John Dalton
(1766 – 1844)

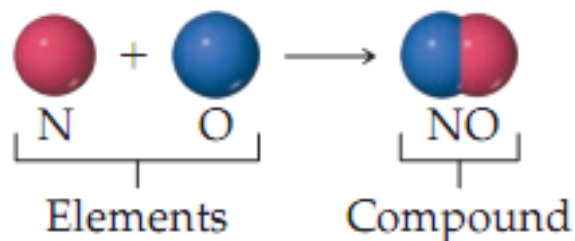
- Each element is composed of small particles called **atoms**.
- All atoms of a given element are **identical**.



- Atoms are **neither created nor destroyed nor changed from one type to another** in chemical reactions,



- **Compounds** are formed when atoms of **more than one element** combine.



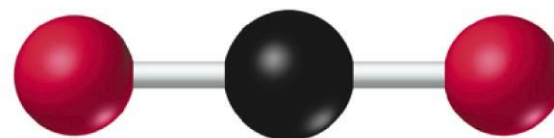
Consequences of Dalton's theory

- ◆ **Law of Definite Proportions:** combinations of elements are in ratios of small whole numbers.

☆ In forming carbon monoxide, 1.33 g of oxygen combines with 1.0 g of carbon.



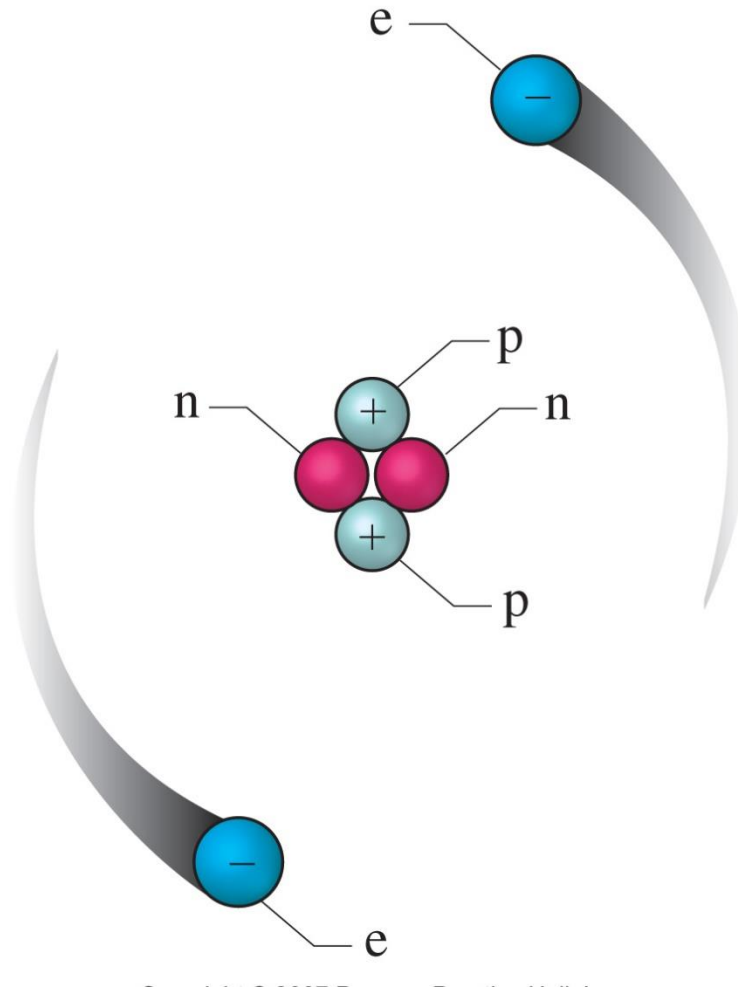
☆ In the formation of hydrogen peroxide 2.66 g of oxygen combines with 1.0 g of hydrogen.



2-2 Discovery in Atomic Structure

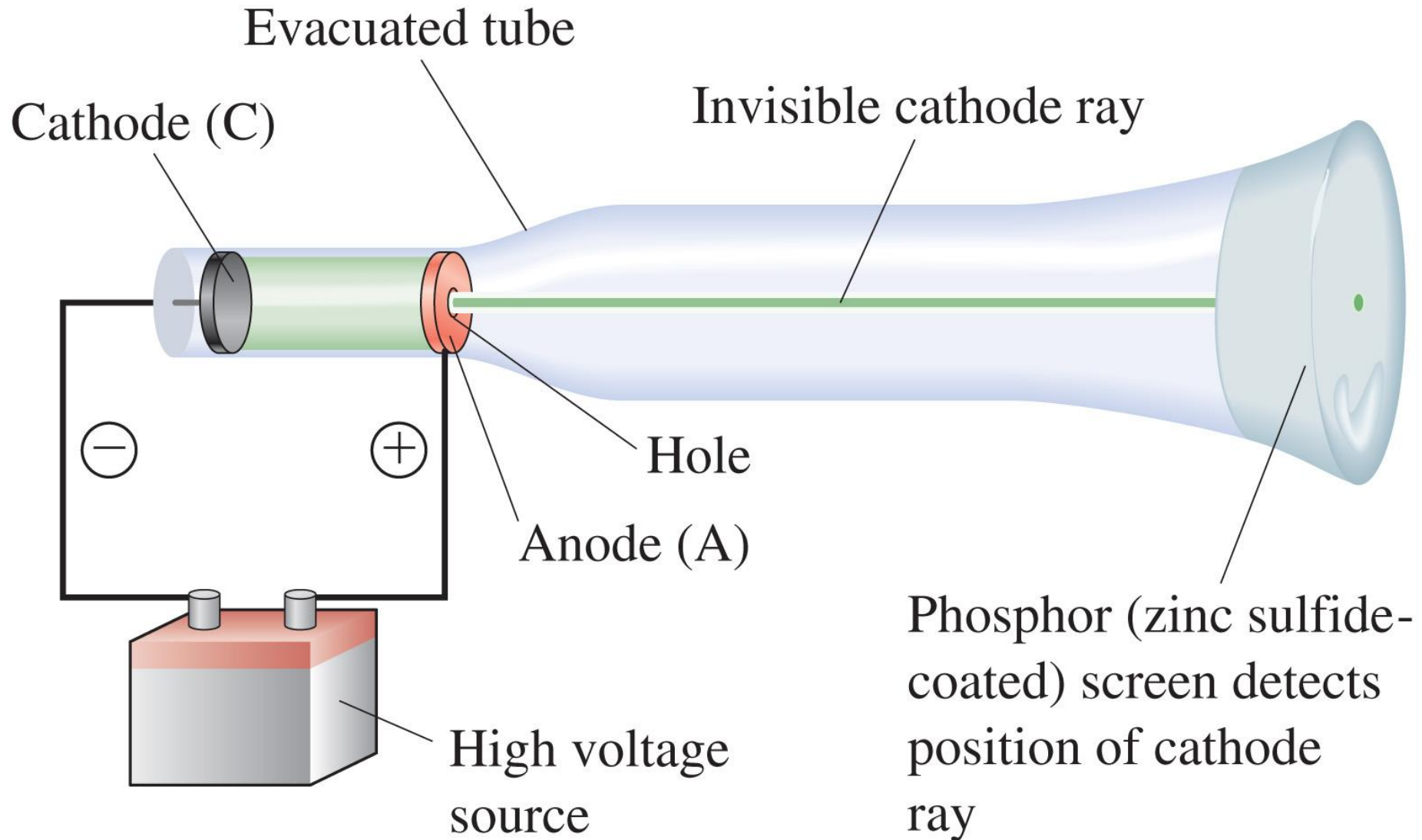
Thomson
electrons 1897

Rutherford
protons 1919

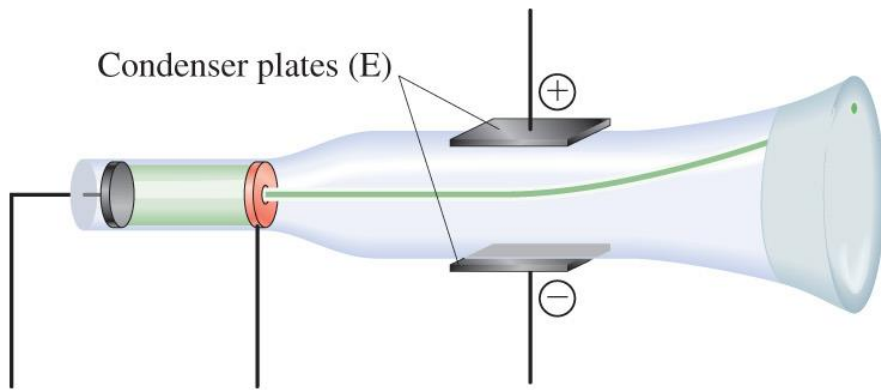


James Chadwick
neutrons 1932

Cathode rays and Electrons

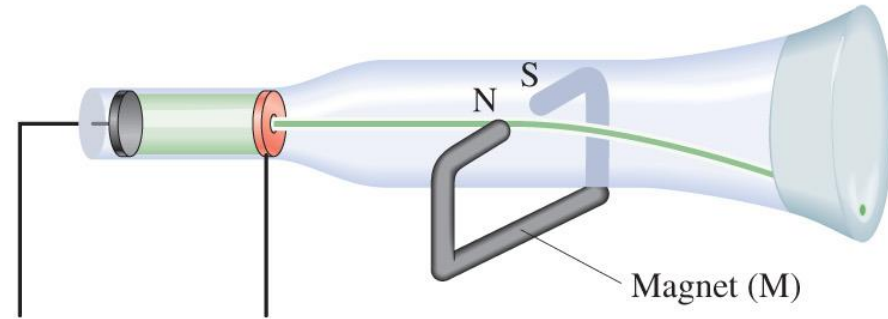


Properties of cathode rays & Discovery of electron



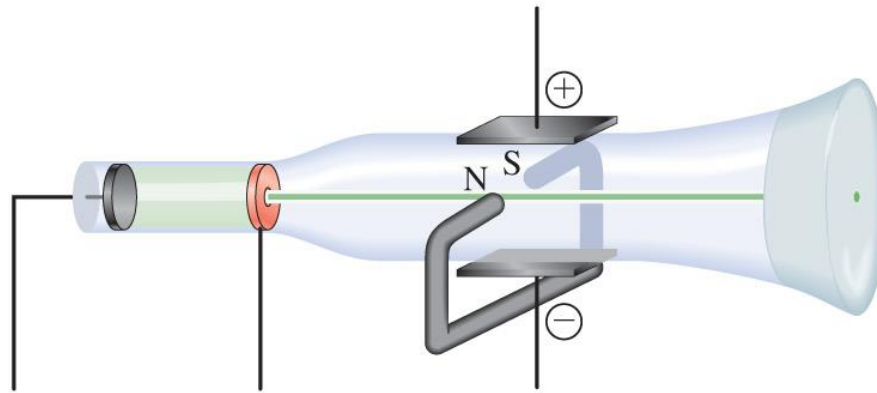
(a)

deflected by an electric field



(b)

deflected by a magnetic field

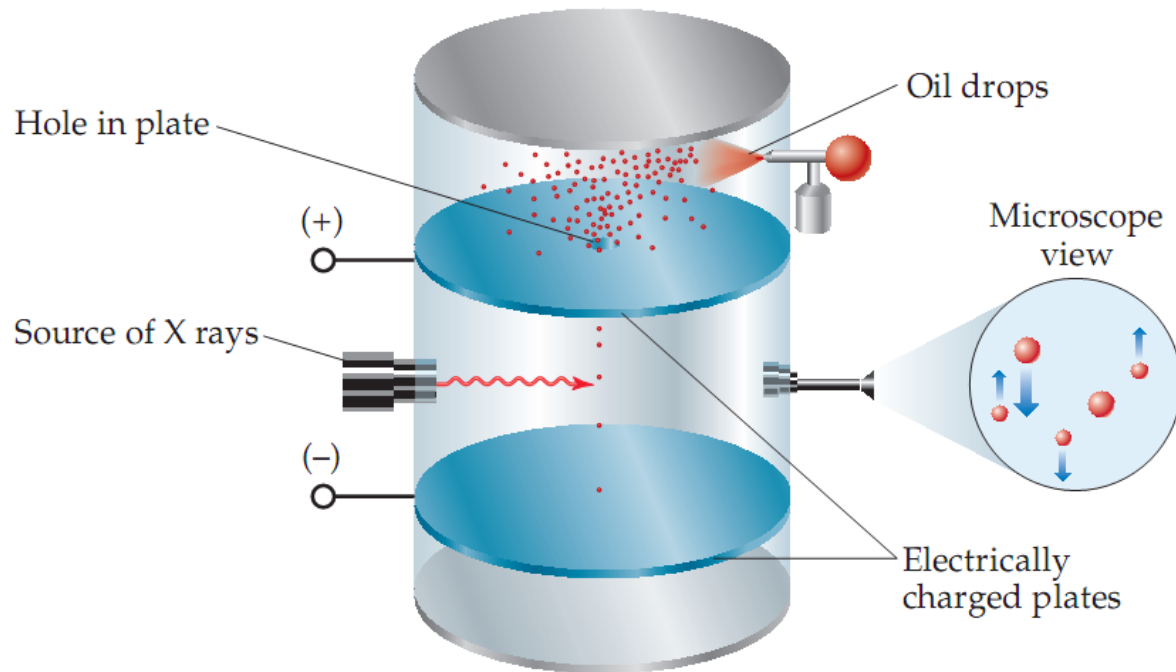


(c)

Thomson's charge-to-mass ratio $e/m = 1.76 \times 10^8$ coulombs/g

Charge on the electron

Would the masses of the oil drops be changed significantly by any electrons that accumulate on them?

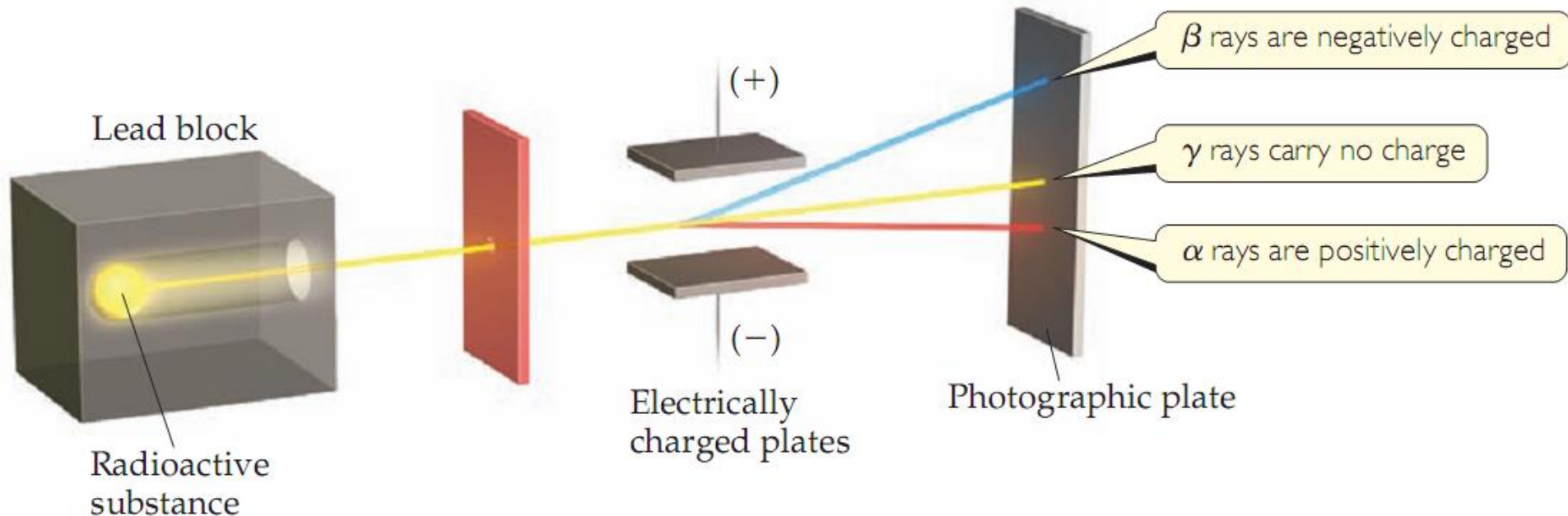


- ★ The ionized oil drops can be balanced against the pull of gravity by an electric field.
- ★ The charge is an *integral* multiple of the electronic charge, $1.602 \times 10^{-19} \text{ C}$.

Mass of the electron

$$\text{Electron mass} = \frac{1.602 \times 10^{-19} \text{ C}}{1.76 \times 10^8 \text{ C/g}} = 9.10 \times 10^{-28} \text{ g}$$

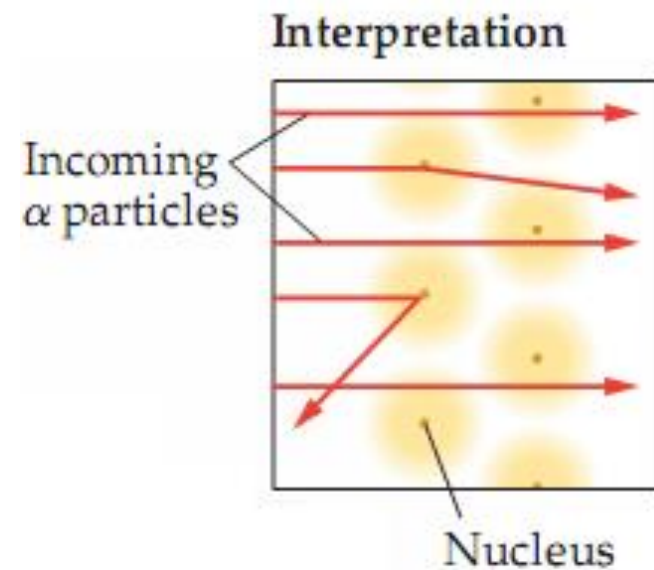
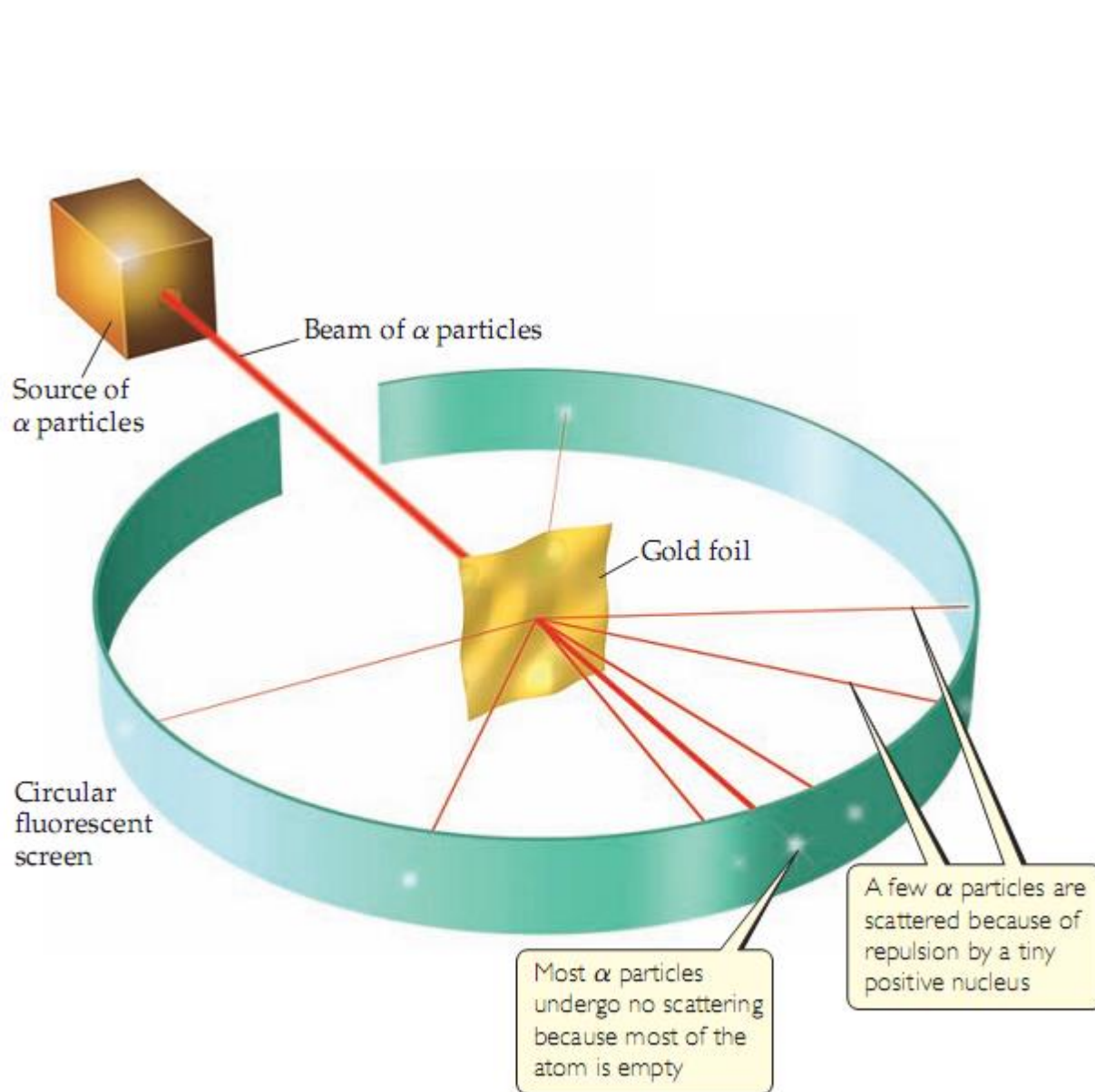
Radioactivity



Radioactivity is the spontaneous emission of radiation from a substance.

- X-rays and γ -rays are high-energy light.
- α -particles are a stream of helium nuclei, He^{2+} .
- β -particles are a stream of high speed electrons that originate in the nucleus.

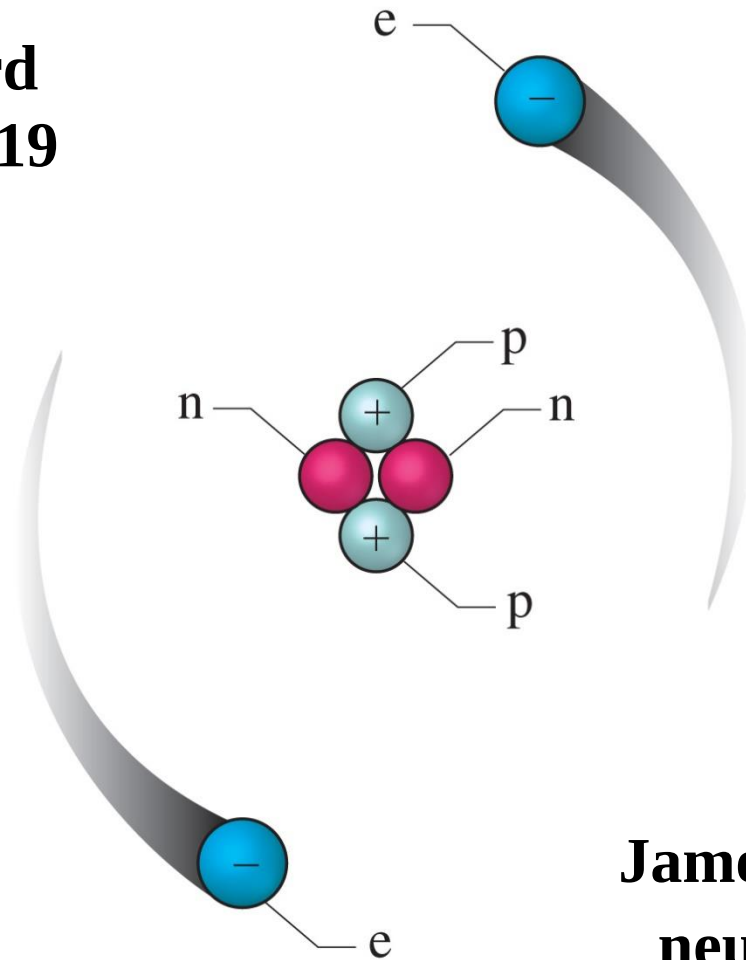
Geiger and Rutherford 1909



Most of the mass and all of the positive charge is concentrated in a small region called the nucleus .

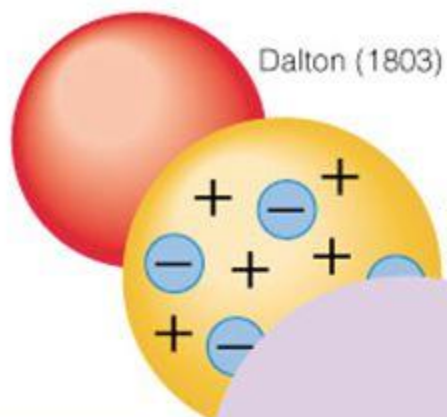
There are as many electrons outside the nucleus as there are units of positive charge on the nucleus

**Rutherford
protons 1919**



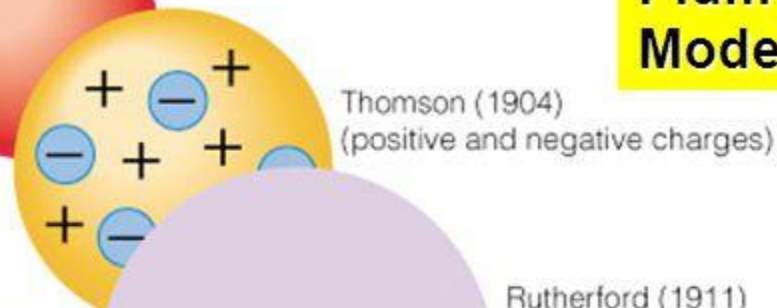
**James Chadwick
neutrons 1932**

Marble Model



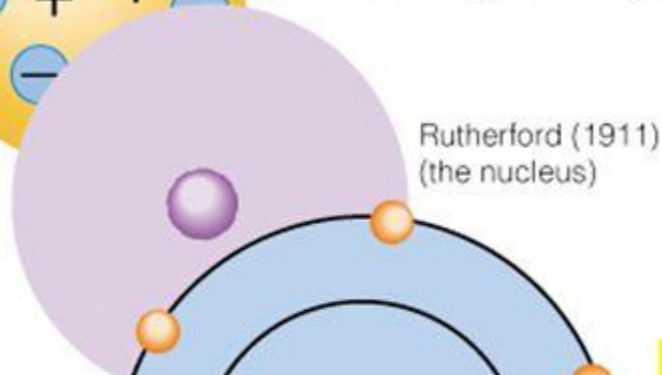
Dalton (1803)

Plum Pudding Model



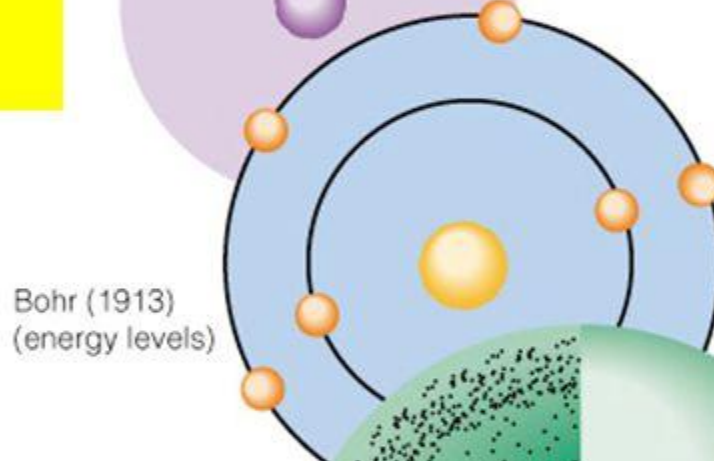
Thomson (1904)
(positive and negative charges)

The Nuclear Model



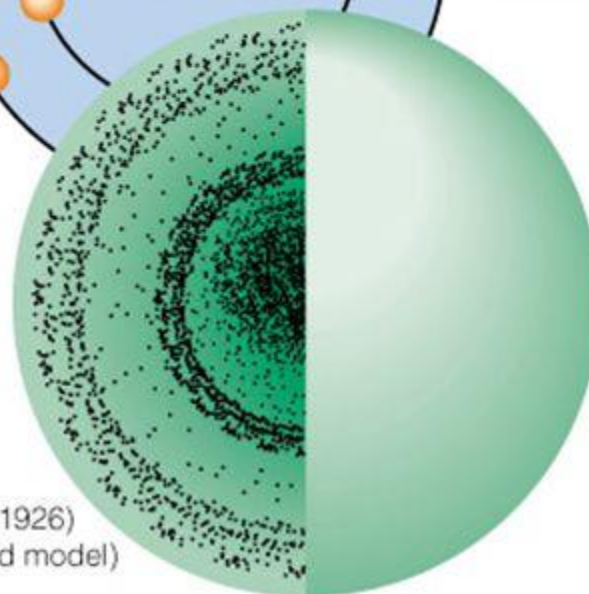
Rutherford (1911)
(the nucleus)

The Planetary Model

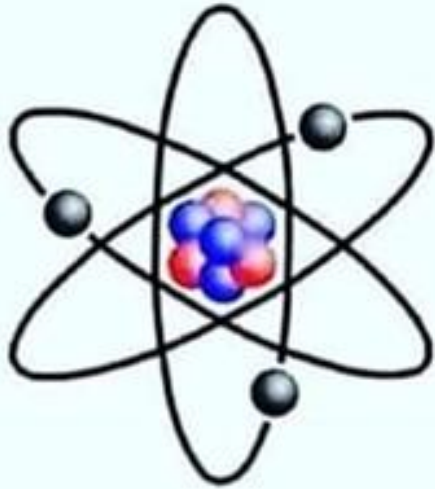


Bohr (1913)
(energy levels)

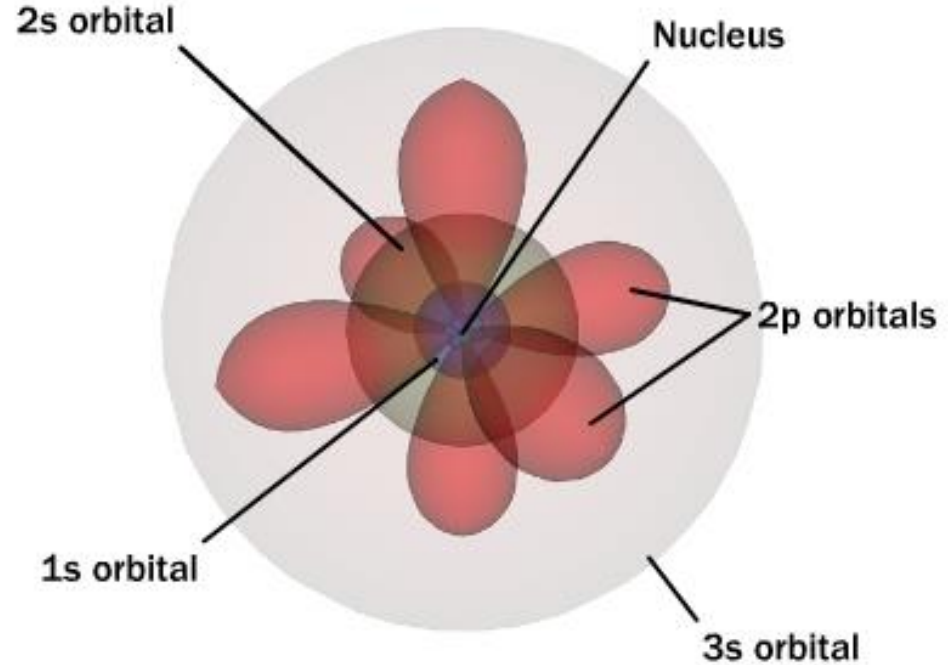
Schrödinger (1926)
(electron cloud model)



**When people still think
that atoms look like this**



instead of this



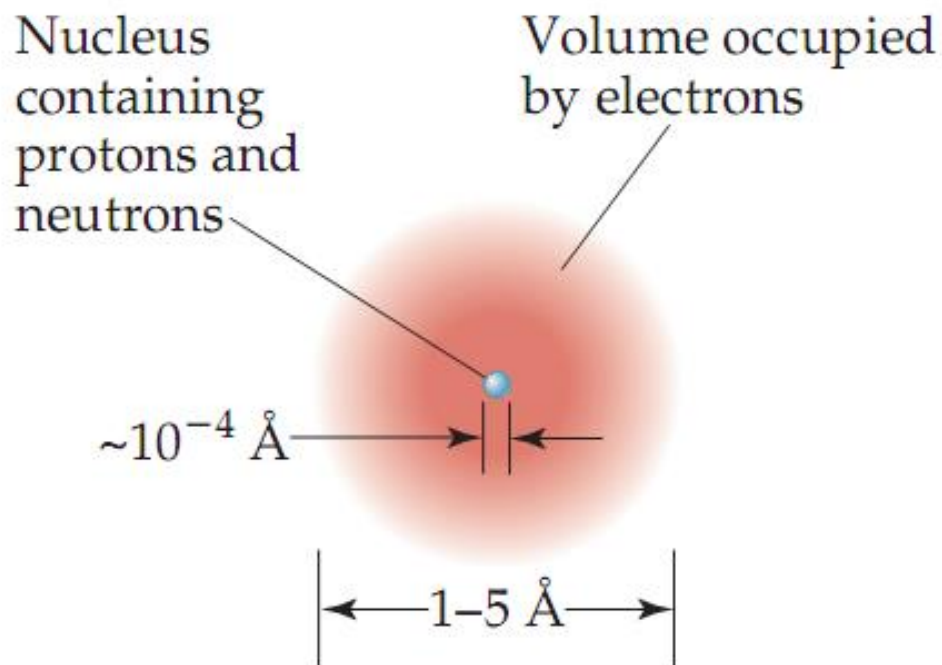
©2001 How Stuff Works



"You disappoint me."

**Werner Heisenberg
(1901-1976)**

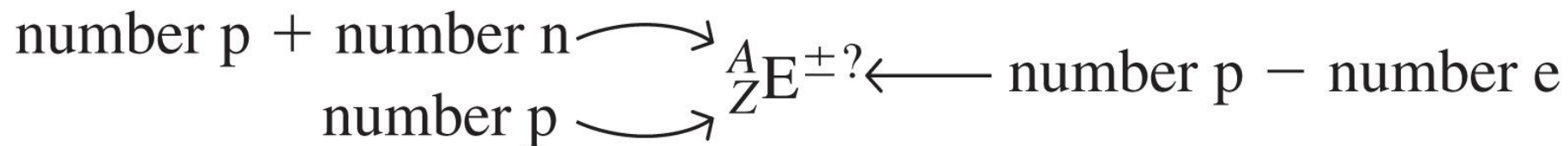
2-3 The Modern View of Atomic Structure



Every atom has an equal number of electrons and protons, so atoms have no net electrical charge.

Chemical Elements

***To represent a particular atom we use symbolism:**



A = mass number

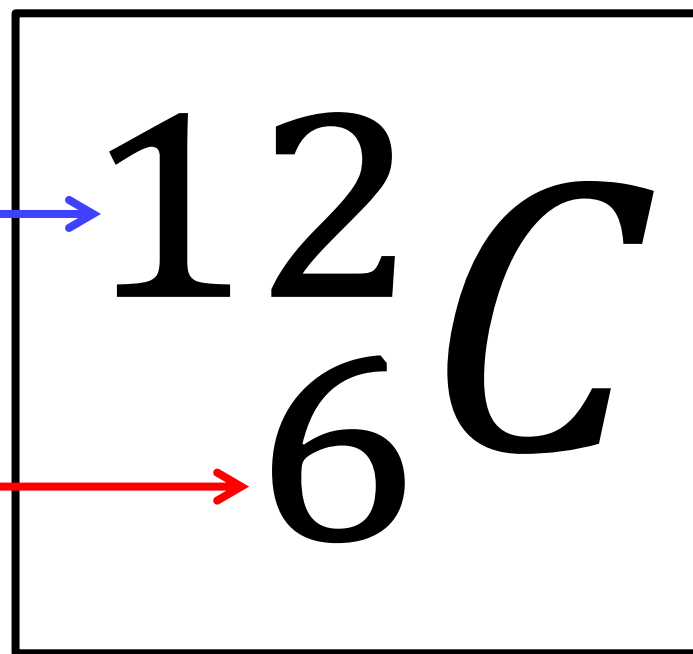
Z = atomic number

Atomic Number & Mass Number

Symbol of carbon

Mass number
(# of protons plus neutrons)

Atomic number
(# of protons)

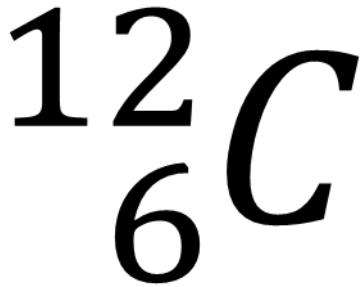


The **atomic number (Z) is always the same** for all atoms with same letter symbol

The **mass number (A) can change** for different atoms with the same letter symbol

Isotopes

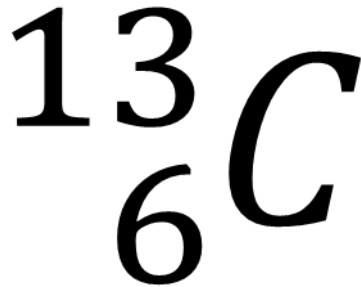
Isotopes are atoms with the same **atomic number** but different atomic masses. This happens when two otherwise identical atoms have different numbers of **neutrons**.



Carbon twelve



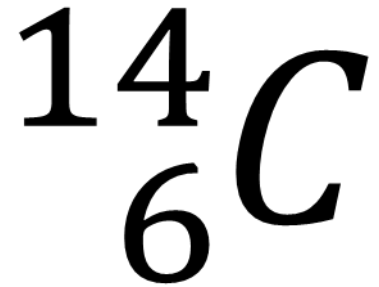
6 protons + 6 neutrons



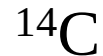
Carbon thirteen



6 protons + 7 neutrons



Carbon fourteen



6 protons + 8 neutrons

2-4 Atomic Weights

***The heaviest atom has a mass of only 4.8×10^{-22} g and a diameter of only 5×10^{-10} m.**

Useful units:

☆ **1 amu (atomic mass unit) = 1.66054×10^{-27} kg**

☆ **1 pm (picometer) = 1×10^{-12} m**

☆ **1 Å (Angstrom) = 1×10^{-10} m = 100 pm = 1×10^{-8} cm**

Biggest atom is 240 amu and is 50 Å across.

Typical C-C bond length 154 pm (1.54 Å)

Molecular models are 1 Å /inch or about 0.4 Å /cm

Atomic mass unit (amu)

one twelfth of the mass of an unbound neutral atom of carbon 12 in its nuclear and electronic ground state and at rest, and has a value of $1.660539040(20) \times 10^{-27}$ kg,

Particle	Mass		Electric Charge	
	kg	amu	Coulombs	(<i>e</i>)
Electron	9.1094×10^{-31}	0.00054858	-1.6022×10^{-19}	-1
Proton	1.6726×10^{-27}	1.0073	$+1.6022 \times 10^{-19}$	+1
Neutron	1.6749×10^{-27}	1.0087	0	0

Atomic Weight vs Atomic Mass

- Most of carbon atoms have just 6 protons and 6 neutrons

$$\text{Atomic mass of } ^{12}\text{C} = 6+6 = 12$$

- Some carbon atoms have 6 protons and 7 neutrons

$$\text{Atomic mass of } ^{13}\text{C} = 6+7 = 13$$

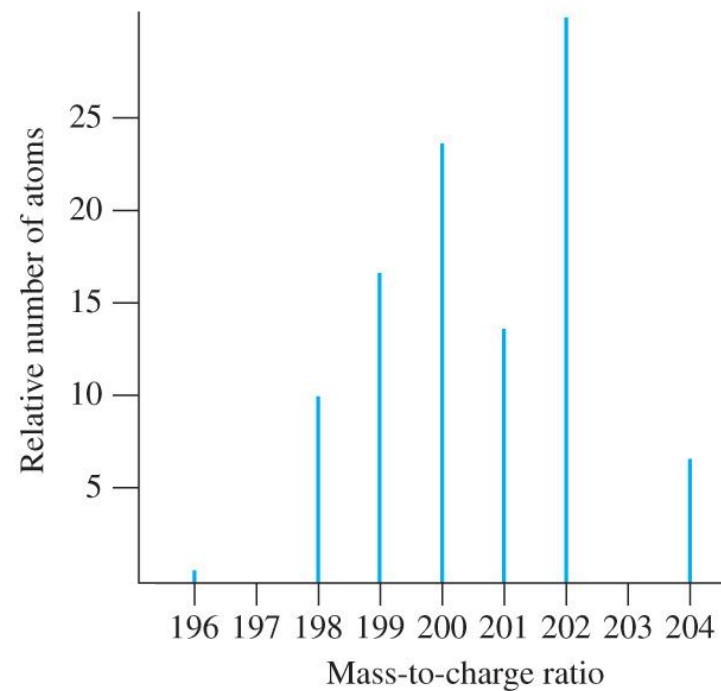
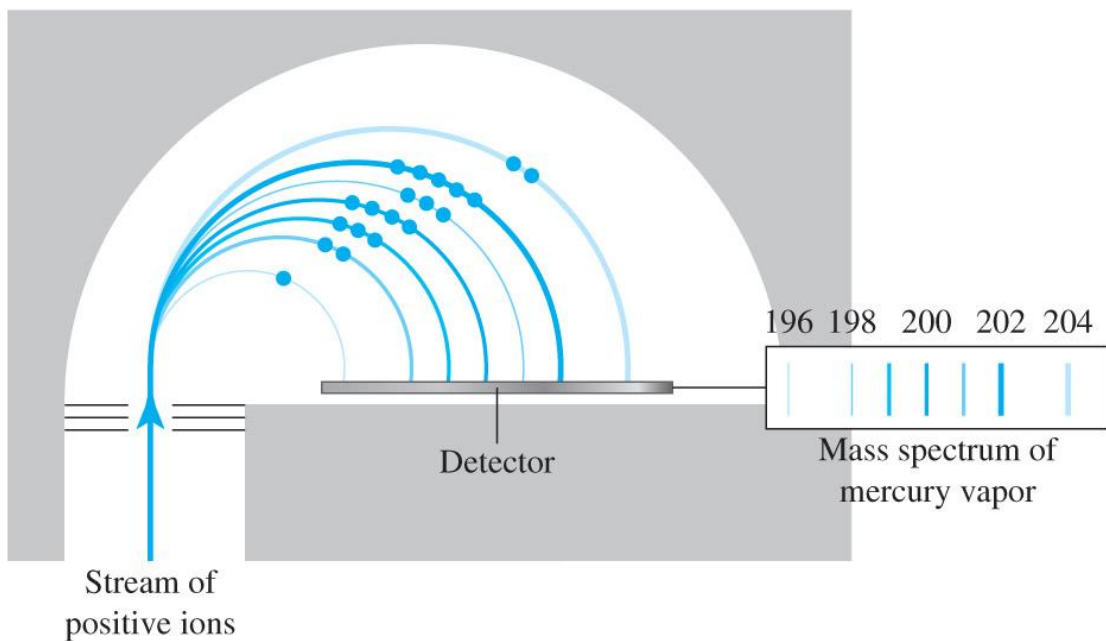
- Even fewer number of carbon atoms have 6 protons and 8 neutrons

$$\text{Atomic mass of } ^{14}\text{C} = 6+8 = 14$$

Which one is most representative?

$$\text{Atomic weight} = \sum_{\substack{\text{over all} \\ \text{isotopes of} \\ \text{the element}}} [(\text{isotope mass}) \times (\text{fractional isotope abundance})]$$

Fractional Isotope Abundance



EXAMPLE

Naturally occurring carbon is composed of 98.93% ^{12}C and 1.07% ^{13}C .

The atomic mass of ^{12}C is (exactly) 12 u, and

the atomic mass of ^{13}C is 13.00335 u,

making the atomic weight of carbon:

$$\left(\frac{98.93}{100}\right)12u + \left(\frac{1.07}{100}\right)13.00335u = 12.01u$$

The sum of the percent natural abundances must be 100%.

EXAMPLE

Naturally occurring chlorine is 75.78% ^{35}Cl (atomic mass 34.969 u) and 24.22% ^{37}Cl (atomic mass 36.966 u). Calculate the atomic weight of chlorine.

$$\begin{aligned}\text{Atomic weight} &= (0.7578)(34.969 \text{ u}) + (0.2422)(36.966 \text{ u}) \\ &= 26.50 \text{ u} + 8.953 \text{ u} \\ &= 35.45 \text{ u}\end{aligned}$$

EXAMPLE

Relating the Masses and Natural Abundances of Isotopes to the Atomic Mass of an Element.

Bromine has two naturally occurring isotopes, bromine-79 and bromine-81. Bromine-79 has a mass of 78.9183 u and an abundance of 50.69%. Use the atomic weight of bromine found in the periodic table to determine the mass and percent of natural abundance of bromine-81?

EXAMPLE

Solution

Write the general equations

$$100\% = \chi_1 + \chi_2 + \chi_3 \dots$$

$$\text{Atomic mass} = \chi_1 \times m_1 + \chi_2 \times m_2 + \chi_3 \times m_3 \dots$$

EXAMPLE

Identify the knowns and unknowns in the specific equations

$$\text{Atomic mass} = \chi_{\text{Br-79}} \times m_{\text{Br-79}} + \chi_{\text{Br-81}} \times m_{\text{Br-81}}$$

Solve

$$100\% = \chi_{79} + \chi_{81}$$

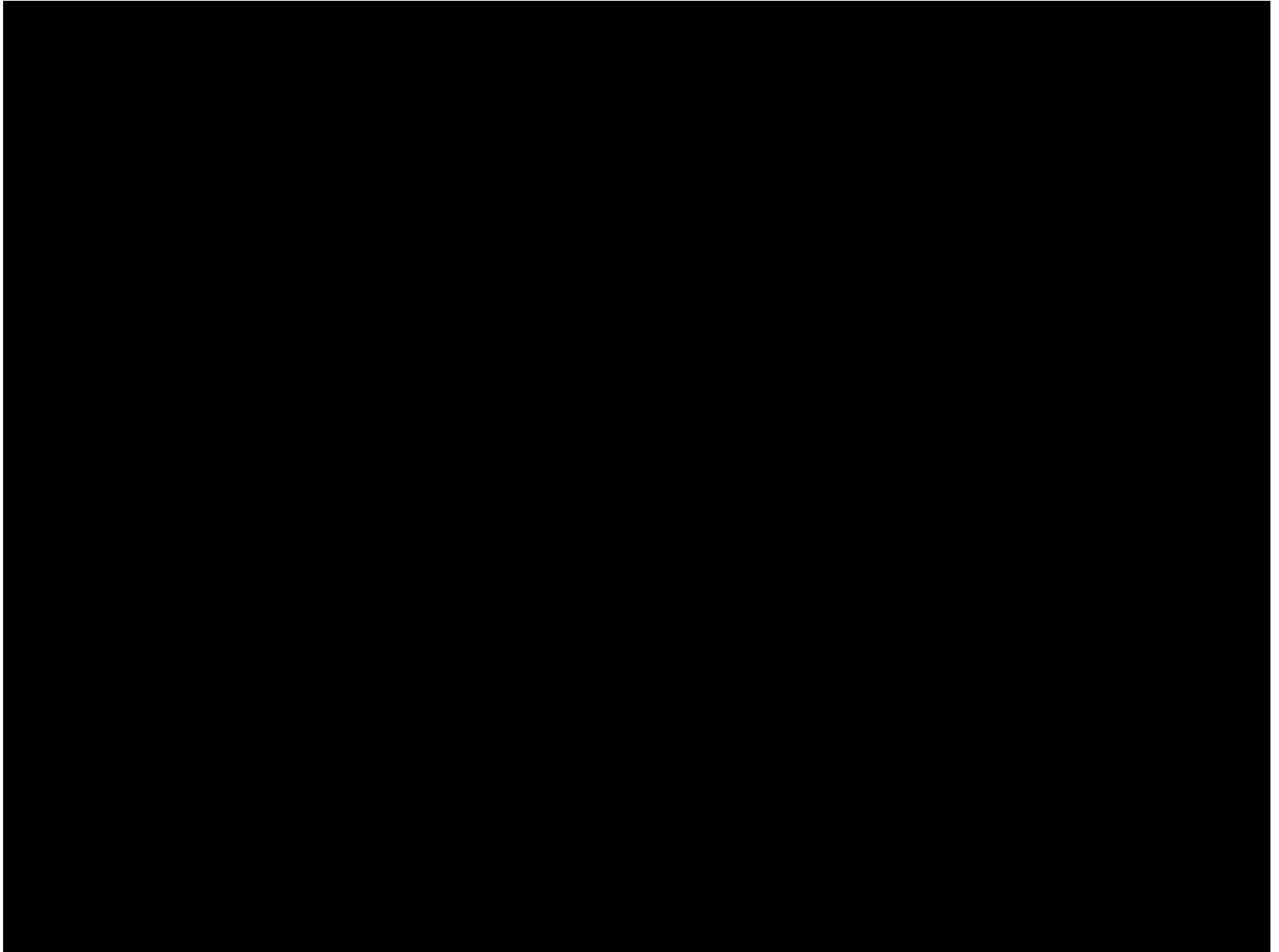
$$\chi_{81} = 100\% - \chi_{79}$$

$$m_{\text{Br-81}} = \frac{\text{Atomic mass} - (\chi_{\text{Br-79}} \times m_{\text{Br-79}})}{\chi_{\text{Br-81}}}$$

Calculate

$$m_{\text{Br-81}} = \frac{79.904 \text{ u} - (0.5069 \times 78.9183 \text{ u})}{0.4931} = 80.92 \text{ u}$$

2-5 The Periodic Table



1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18

Periodic Table of Elements																		VIII A											
IA		IIA		Metals										Nonmetals		VIII A													
				Alkali metals		Alkaline earth metals		Lanthanoids (Lanthanides)		Actinoids (Actinides)		Transition metals		Post-transition metals		Metalloids		Other nonmetals		Noble gases									
C Solid		Hg Liquid		H Gas		Rf Unknown																							
1		3		4																		2							
H		Li		Be																		He							
Hydrogen 1.008		Lithium 6.94		Beryllium 9.0122																		Helium 4.0026							
2		11		12																		10							
		Na		Mg																		Ne							
		Sodium 22.990		Magnesium 24.305																		Neon 20.180							
3		19		20		21		22		23		24		25		26		27		28		30							
		K		Ca		Sc		Ti		V		Cr		Mn		Fe		Co		Ni		Zn							
		Potassium 39.098		Calcium 40.078		Scandium 44.956		Titanium 47.867		Vanadium 50.942		Chromium 51.996		Manganese 54.938		Iron 55.845		Cobalt 58.933		Nickel 58.693		Copper 63.546							
4		37		38		39		40		41		42		43		44		45		46		48							
		Rb		Sr		Y		Zr		Nb		Mo		Tc		Ru		Rh		Pd		Cd							
		Rubidium 85.468		Strontium 87.62		Yttrium 88.906		Zirconium 91.224		Niobium 92.906		Molybdenum 95.95		Technetium (98)		Ruthenium 101.07		Rhodium 102.91		Palladium 106.42		Cadmium 112.41							
5		55		56		57-71		72		73		74		75		76		77		78		80							
		Cs		Ba				Hf		Ta		W		Re		Os		Ir		Pt		Au							
		Caesium 132.91		Barium 137.33				Hafnium 178.49		Tantalum 180.95		Tungsten 183.84		Rhenium 186.21		Osmium 190.23		Iridium 192.22		Platinum 195.08		Gold 196.97							
6		87		88		89-103		104		105		106		107		108		109		110		111							
		Fr		Ra				Rf		Db		Sg		Bh		Hs		Mt		Ds		Rg							
		Francium (223)		Radium (226)				Rutherfordium (267)		Dubnium (268)		Seaborgium (269)		Bohrium (270)		Hassium (277)		Meitnerium (278)		Darmstadtium (281)		Roentgenium (282)							
7		113		114		115		116		117		118		119		120		121		122		123							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		13		14		15		16		17		18		19		20		21		22		23							
		Al		Si		P		S		Cl		Ar		K		Ca		Sc		Ti		V							
		Aluminium 26.982		Silicon 28.085		Phosphorus 30.974		Sulfur 32.06		Chlorine 35.45		Argon 39.948		Potassium 39.098		Calcium 40.078		Scandium 44.956		Titanium 47.867		Vanadium 50.942							
		31		32		33		34		35		36		37		38		39		40		41							
		Ga		Ge		As		Se		Br		Kr		Rb		Sr		Y		Zr		Nb							
		Gallium 69.723		Germanium 72.630		Arsenic 74.922		Selenium 78.971		Bromine 79.904		Krypton 83.798		Rubidium 85.468		Strontium 87.62		Yttrium 88.906		Zirconium 91.224		Niobium 92.906							
		49		50		51		52		53		54		55		56		57-71		72		73							
		In		Sn		Sb		Te		I		Xe		Cs		Ba				Hf		Ta							
		Indium 114.82		Tin 118.71		Antimony 121.76		Tellurium 127.60		Iodine 126.90		Xenon 131.29		Caesium 132.91		Barium 137.33				Hafnium 178.49		Tantalum 180.95							
		81		82		83		84		85		86		87		88		89-103		104		105							
		Tl		Pb		Bi		Po		At		Rn		Fr		Ra				Rf		Db							
		Thallium 204.38		Lead 207.2		Bismuth 208.98		Polonium (209)		Astatine (210)		Radon (222)		Francium (223)		Radium (226)				Rutherfordium (267)		Dubnium (268)							
		101		102		103		104		105		106		107		108		109		110		111							
		Md		No		Lr																							
		Mendelevium (258)		Nobelium (259)		Lawrencium (260)																							
		109		110		111		112		113		114		115		116		117		118		119							
		Mt		Ds		Rg		Cn		Nh		Fl		Mc		Lv		Ts		Og									
		Meitnerium (278)		Darmstadtium (281)		Roentgenium (282)		Copernicium (285)		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)									
		121		122		123		124		125		126		127		128		129		130		131							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		131		132		133		134		135		136		137		138		139		140		141							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		141		142		143		144		145		146		147		148		149		150		151							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		151		152		153		154		155		156		157		158		159		160		161							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		161		162		163		164		165		166		167		168		169		170		171							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		171		172		173		174		175		176		177		178		179		180		181							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		181		182		183		184		185		186		187		188		189		190		191							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		191		192		193		194		195		196		197		198		199		200		201							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		201		202		203		204		205		206		207		208		209		210		211							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		211		212		213		214		215		216		217		218		219		220		221							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		221		222		223		224		225		226		227		228		229		230		231							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		231		232		233		234		235		236		237		238		239		240		241							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		241		242		243		244		245		246		247		248		249		250		251							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		251		252		253		254		255		256		257		258		259		260		261							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		261		262		263		264		265		266		267		268		269		270		271							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		271		272		273		274		275		276		277		278		279		280		281							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		281		282		283		284		285		286		287		288		289		290		291							
		Nh		Fl		Mc		Lv		Ts		Og																	
		Nihonium (286)		Flerovium (289)		Moscovium (290)		Livermorium (293)		Tennessine (294)		Oganesson (294)																	
		291		292		293		294		295		296		297		298		299		300		301							
		Nh		Fl		Mc		Lv		Ts		Og																	

The Long Periodic table

Metals					Nonmetals	
Alkali metals	Alkaline earth metals	Lanthanoids (Lanthanides)	Actinoids (Actinides)	Transition metals	Post-transition metals	Metalloids
						Other nonmetals
						Noble gases

s block

p block

C Solid
 Hg Liquid
 H Gas
 Rf Unknown

f block

d block

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

1

2

C

Solid

Hg

Liquid

H

Gas

Rf

Unknown

1

2

3

4

11

12

19

20

37

38

55

56

87

88

Atomic Symbol

Name

Weight

Hydrogen

1.008

Lithium

6.94

Sodium

22.990

Potassium

39.098

Rubidium

85.468

Cesium

132.91

Francium

(223)

Beryllium

9.0122

Magnesium

24.305

Calcium

40.078

Strontium

87.62

Ytterbium

173.05

Barium

137.33

Radium

(226)

3

4

5

6

7

8

9

10

11

12

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

1

2

3

4

11

12

19

20

37

38

55

56

87

88

Atomic Symbol

Name

Weight

Hydrogen

1.008

Lithium

6.94

Sodium

22.990

Potassium

39.098

Rubidium

85.468

Cesium

132.91

Francium

(223)

Beryllium

9.0122

Magnesium

24.305

Calcium

40.078

Strontium

87.62

Ytterbium

173.05

Barium

137.33

Radium

(226)

3

4

5

6

7

8

9

10

11

12

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

1

2

3

4

11

12

19

20

37

38

55

56

87

88

Atomic Symbol

Name

Weight

Hydrogen

1.008

Lithium

6.94

Sodium

22.990

Potassium

39.098

Rubidium

85.468

Cesium

132.91

Francium

(223)

Beryllium

9.0122

Magnesium

24.305

Calcium

40.078

Strontium

87.62

Ytterbium

173.05

Barium

137.33

Radium

(226)

3

4

5

6

7

8

9

10

11

12

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

1

2

3

4

11

12

19

20

37

38

55

56

87

88

Atomic Symbol

Name

Weight

Hydrogen

1.008

Lithium

6.94

Sodium

22.990

Potassium

39.098

Rubidium

85.468

Cesium

132.91

Francium

(223)

Beryllium

9.0122

Magnesium

24.305

Calcium

40.078

Strontium

87.62

Ytterbium

173.05

Barium

137.33

Radium

(226)

3

4

5

6

7

8

9

10

11

12

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

1

2

3

4

11

12

19

20

37

38

55

56

87

88

Atomic Symbol

Name

Weight

Hydrogen

1.008

Lithium

6.94

Sodium

22.990

Potassium

39.098

Rubidium

85.468

Cesium

132.91

Francium

(223)

Beryllium

9.0122

Magnesium

24.305

Calcium

40.078

Strontium

87.62

Ytterbium

173.05

Barium

137.33

Radium

(226)

3

4

5

6

7

8

9

10

11

12

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

1

2

3

4

11

12

19

20

37

38

55

56

87

88

Atomic Symbol

Name

Weight

Hydrogen

1.008

Lithium

6.94

Sodium

22.990

Potassium

39.098

Rubidium

85.468

Cesium

132.91

Francium

(223)

Beryllium

9.0122

Magnesium

24.305

Calcium

40.078

Strontium

87.62

Ytterbium

173.05

Barium

137.33

Radium

(226)

3

4

5

6

7

8

9

10

11

12

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
1	1 H Hydrogen 1.008	Atomic # Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids (Lanthanides) Actinoids (Actinides) Transition metals Post-transition metals Metalloids Nonmetals Other nonmetals Noble gases																2 He Helium 4.0026	
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122	Prictogens Chalcogens Halogens																10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305	13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948											
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798	
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29	
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)	
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)	
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																			
6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97				
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)				

- Read atomic masses.
- Read the ions formed by main group elements.
- Read the electron configuration.
- Learn trends in physical and chemical properties.

EXAMPLE

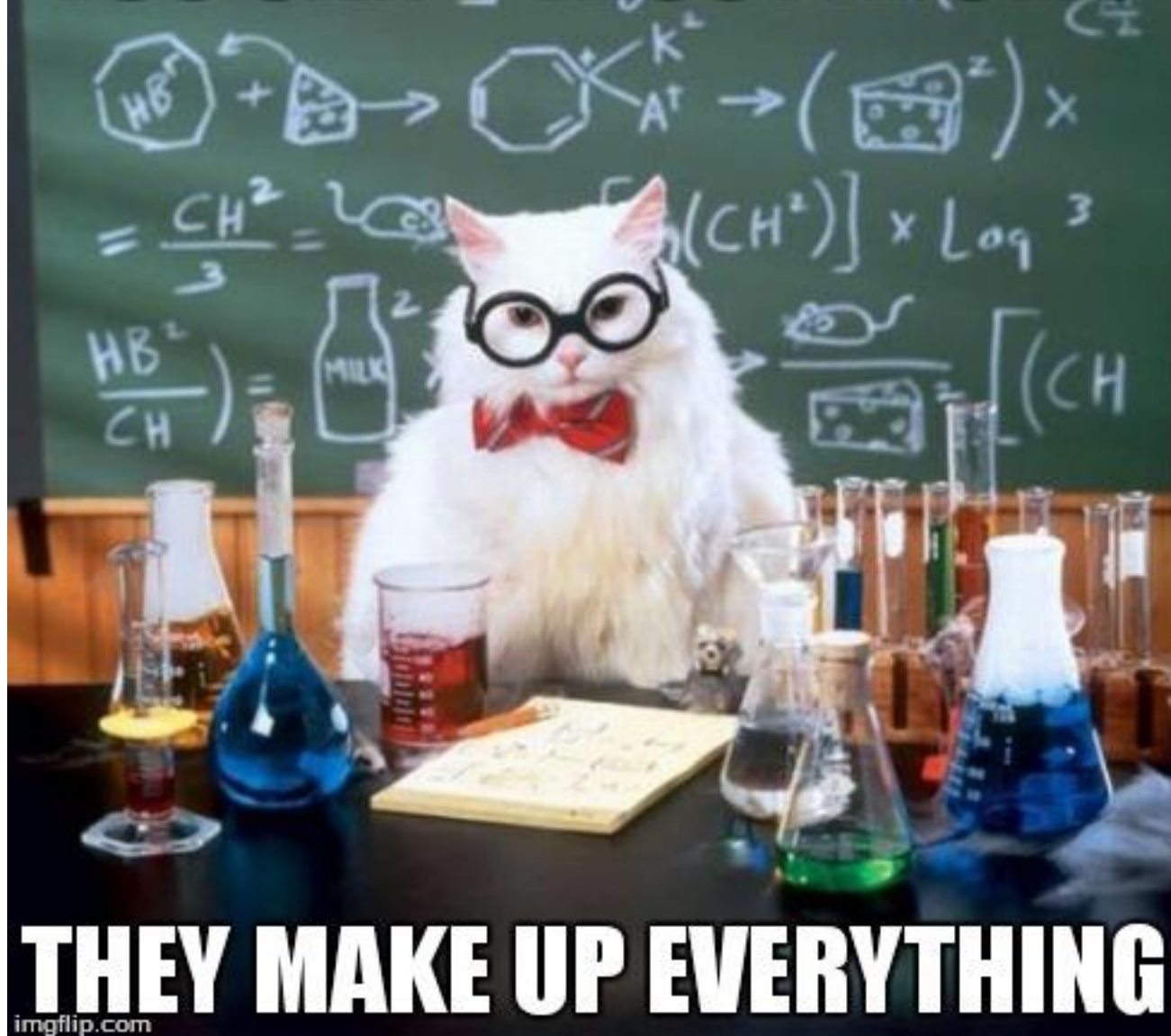
1. Which two of these elements would you expect to show the greatest similarity in chemical and physical properties: B, Ca, F, He, Mg, P?

Your answer: _____

2. A biochemist who is studying the properties of certain sulfur (S)–containing compounds in the body wonders whether trace amounts of another nonmetallic element might have similar behavior. To which element should she turn her attention?

- (a) O,
- (b) As,
- (c) Se,
- (d) Cr,
- (e) P

YOU CAN'T TRUST ATOMS



2-6 Molecules and Molecular Compounds

- Only the noble-gas elements are normally found in nature as isolated atoms.
- Most matter is composed of molecules or ions.

Diatomic molecules



Hydrogen, H_2



Oxygen, O_2

Molecular compounds



Water, H_2O



Methane, CH_4



Carbon
monoxide, CO



Ethylene, C_2H_4

Molecular compounds are generally composed of nonmetals only

Molecular and Empirical Formulas

- Chemical formulas that indicate the **actual numbers of atoms** in a molecule are called **molecular formulas**
- Chemical formulas that give only the **relative number of atoms of each type** in a molecule are called **empirical formulas**

Example:

	Hydrogen peroxide	Ethylene
Molecular formula	H_2O_2	C_2H_4
Empirical formula	HO	CH_2

- For many substances, the molecular formula and the empirical formula are identical, as in the case of water, H_2O

EXAMPLE

Write the empirical formulas for

- (a) glucose, a substance also known as either blood sugar or dextrose—molecular, whose molecular formula is $\text{C}_6\text{H}_{12}\text{O}_6$;
- (b) nitrous oxide, a substance used as an anesthetic and commonly called laughing gas—molecular, whose molecular formula is N_2O
- (c) decaborane, whose molecular formula is $\text{B}_{10}\text{H}_{14}$

PRACTICE 1

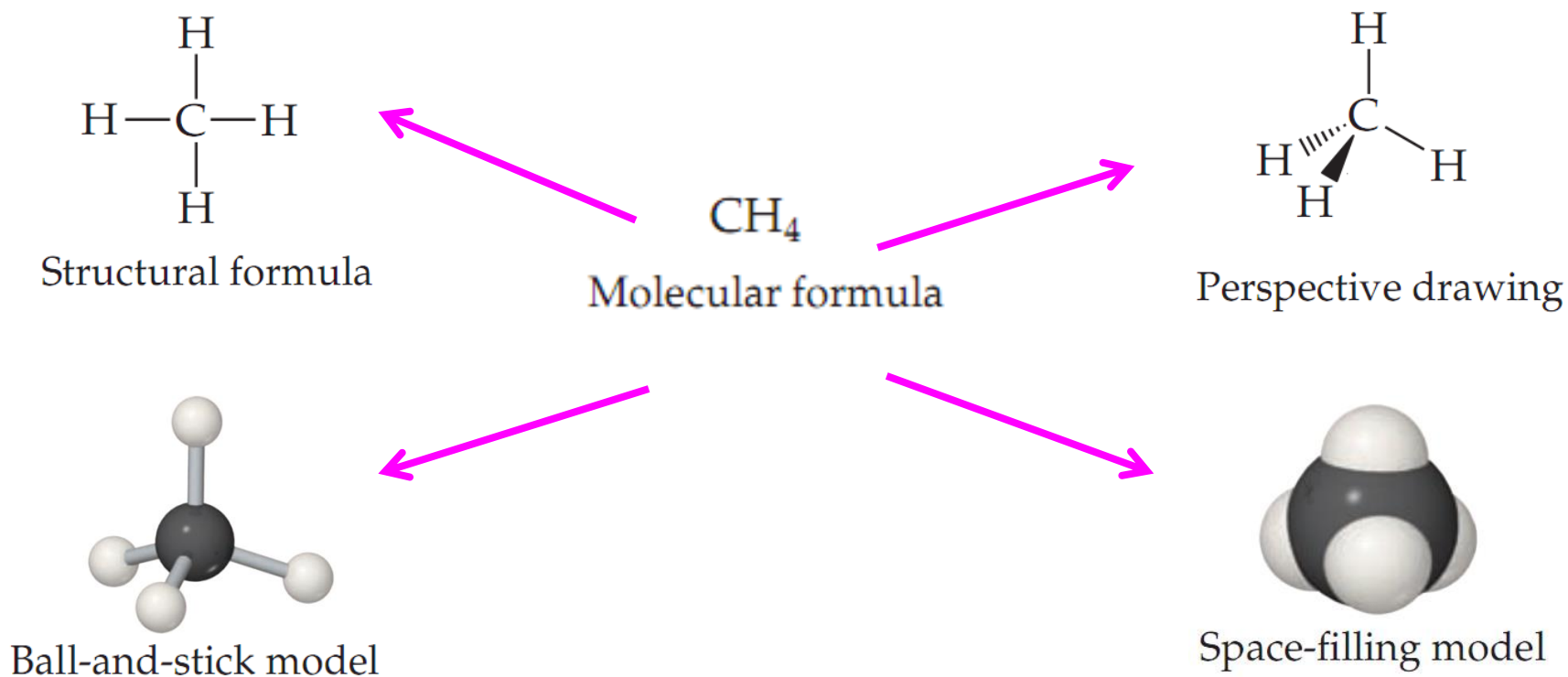
Tetracarbon dioxide is an unstable oxide of carbon with the following molecular structure:



What are the molecular and empirical formulas of this substance?

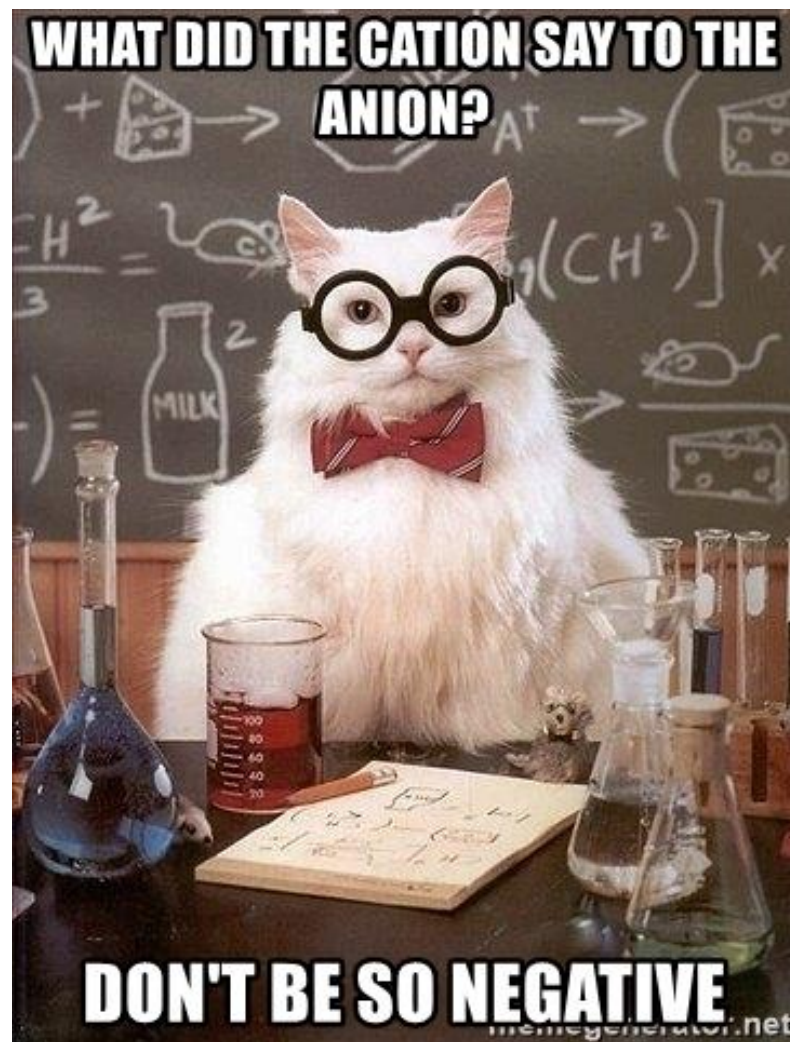
- (a) C_2O_2 , CO_2 ,
- (b) C_4O , CO ,
- (c) CO_2 , CO_2 ,
- (d) C_4O_2 , C_2O ,
- (e) C_2O , CO_2 .

Picturing Molecules

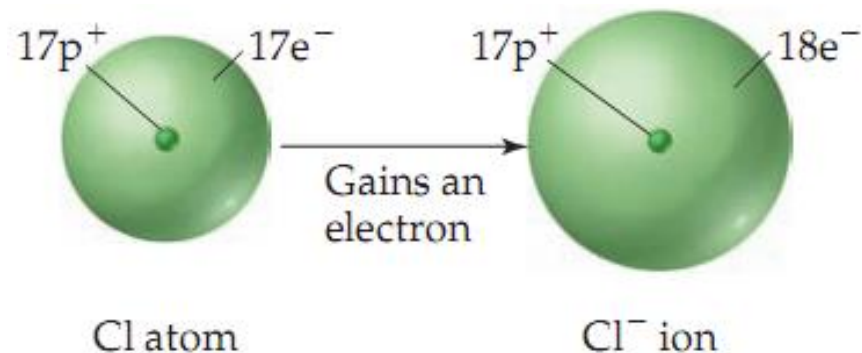
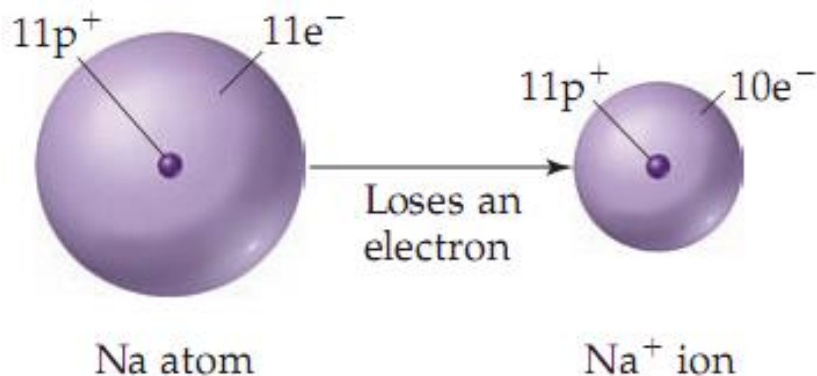


- A **symbol-and-line formula** usually does not show the actual angles between atoms
- A **perspective drawing** shows the three-dimensional shape
- A **ball-and-stick model** represents the angles between atoms
- A **space-filling model** show the relative sizes of the atoms, but the angles between atoms

2-7 Ions and Ionic Compounds



How do ions form?



- A **positive** charge ion is a **cation** (pronounced CAT-ion);
- A **negatively** charged ion is an **anion** (AN-ion)

EXAMPLE

Give the chemical symbol, including superscript indicating mass number, for

(a) the ion with 22 protons, 26 neutrons, and 19 electrons;

(b) the ion of sulfur that has 16 neutrons and 18 electrons.

EXAMPLE

1. In which of the following species is the number of protons less than the number of electrons?

(a) Ti^{2+} ,

(b) P^{3-} ,

(c) Mn ,

(d) Se_4^{2-}

(e) Ce^{4+} .

Your answer: _____

2. How many protons, neutrons, and electrons does the $^{79}\text{Se}^{2-}$ ion possess?

Your answer: _____ protons, _____ neutrons, _____ electrons

Predicting Ionic Charges

- The noble gases are chemically nonreactive elements that form very few compounds.
- Many atoms gain or lose electrons to end up with **the same number of electrons as the noble gas closest to them** in the periodic table.

1A	2A	Transition metals						3A	4A	5A	6A	7A	8A
H ⁺										N ³⁻	O ²⁻	F ⁻	H ⁻
Li ⁺								Al ³⁺			S ²⁻	Cl ⁻	
Na ⁺	Mg ²⁺										Se ²⁻	Br ⁻	
K ⁺	Ca ²⁺										Te ²⁻	I ⁻	
Rb ⁺	Sr ²⁺												
Cs ⁺	Ba ²⁺												

N
O
B
L
E

G
A
S
E
S

- The red stepped line that divides metals from nonmetals also separates cations from anions.*
- Hydrogen forms both 1+ and 1- ions.*

EXAMPLE

1. Predict the charge expected for the most stable ion of

(a) barium

(b) Oxygen

(c) Aluminum

(d) fluorine.

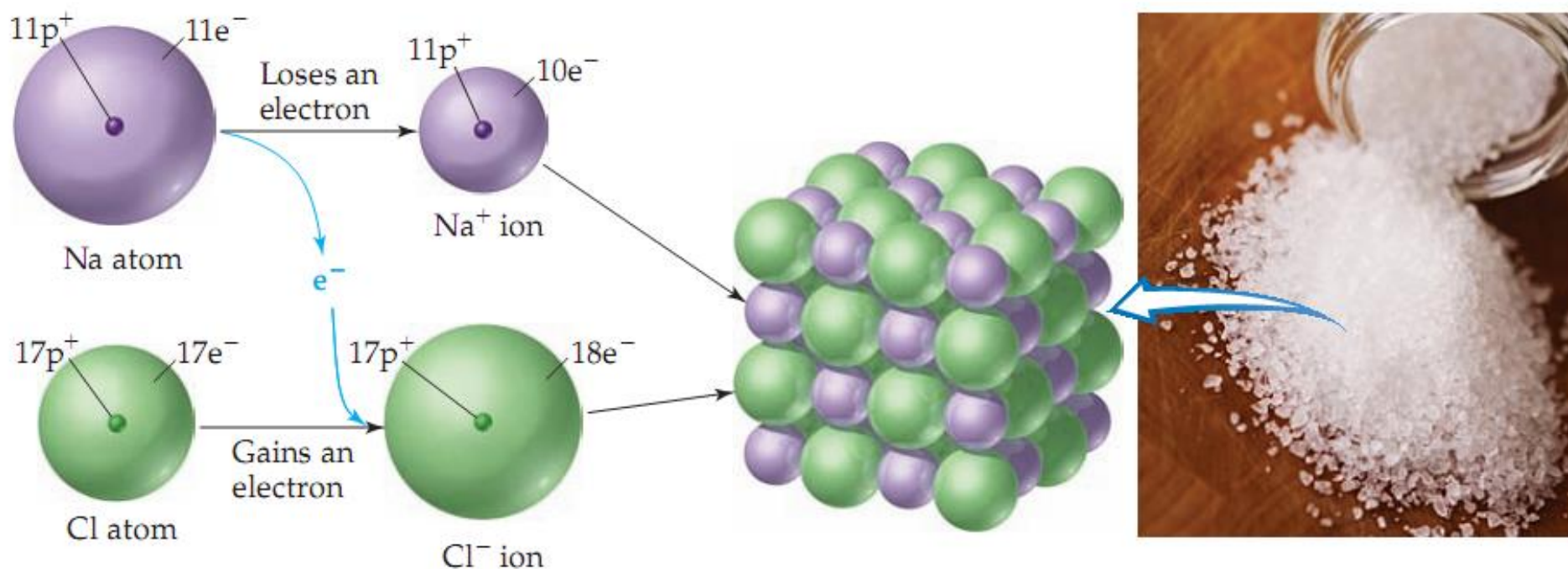
EXAMPLE

3. Use the periodic table to determine which of the following ions has a noble–gas electron arrangement, and which do not. For those that do, indicate the noble–gas arrangement they match:



Ionic Compounds

- A great deal of chemical activity involves the transfer of electrons from one substance to another



Ionic compounds are generally combinations of metals and nonmetals

EXAMPLE

1. Which of these compounds would you expect to be ionic: N_2O , Na_2O , $CaCl_2$, SF_4 ?

Your answer: _____

2. Which of these compounds are molecular: CBr_4 , FeS , P_4O_6 , PbF_2

Your answer: _____

EXAMPLE

Give a reason why each of the following statements is a safe prediction:

- (a) *Every compound of Rb with a nonmetal is ionic in character.*
- (b) *Every compound of nitrogen with a halogen element is a molecular compound.*
- (c) *The compound MgKr_2 does not exist.*
- (d) *Na and K are very similar in the compounds they form with nonmetals.*
- (e) *If contained in an ionic compound, calcium (Ca) will be in the form of the doubly charged ion, Ca^{2+} .*

EXAMPLE

Write the empirical formula of the compound formed by

(a) Al^{3+} and Cl^- ions,

(b) Al^{3+} and O^{2-} ions,

(c) Mg^{2+} and NO_3^- ions.

2-8 Naming Inorganic Compounds



Different ions of the same element have different properties

Naming Cation

a. Cations formed from metal atoms have the same name as the metal:

Na^+ sodium ion

Zn^{2+} zinc ion

Al^{3+} aluminum ion

b. If a metal can form cations with different charges, the positive charge is indicated by a Roman numeral in parentheses following the name of the metal:

Fe^{2+} iron(II) ion (ferrous ion)

Cu^+ copper(I) ion (cuprous ion)

Fe^{3+} iron(III) ion (ferric ion)

Cu^{2+} copper(II) ion (cupric ion)

c. Cations formed from nonmetal atoms have names that end in ***-ium***:

NH_4^+ ammonium ion

H_3O^+ hydronium ion

Naming Anion

- a. The names of monatomic anions are formed by replacing the ending of the name of the element with ***-ide***

H^- hydride ion

O^{2-} oxide ion

N^{3-} nitride ion

A few polyatomic anions also have names ending in ***-ide***

OH^- hydroxide ion

CN^- cyanide ion

O_2^{2-} peroxide ion

Naming Anion (*cont.*)

b. Polyatomic anions containing oxygen are called **oxyanions**,
subfixes

-ate for the most common or representative oxyanion of an element

-ite for an oxyanion that has the same charge but one O atom fewer

NO_3^-	nitrate ion	SO_4^{2-}	sulfate ion
NO_2^-	nitrite ion	SO_3^{2-}	sulfite ion

prefixes

per- indicates one more O atom than the oxyanion ending in **-ate**

hypo- indicates one O atom fewer than the oxyanion ending in **-ite**

ClO_4^-	perchlorate ion (one more O atom than chlorate)
ClO_3^-	chlorate ion
ClO_2^-	chlorite ion (one O atom fewer than chlorate)
ClO^-	hypochlorite ion (one O atom fewer than chlorite)

Naming Ionic Compounds

Names of ionic compounds consist of the cation name followed by the anion name



calcium chloride



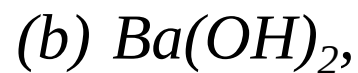
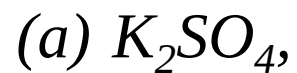
aluminum nitrate



copper(II) perchlorate (or cupric perchlorate)

EXAMPLE

Name the ionic compounds



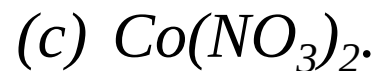
EXAMPLE

*Which of the following ionic compounds is **incorrectly** named?*

- (a) $\text{Zn}(\text{NO}_3)_2$, zinc nitrate;
- (b) TeCl_4 , tellurium(IV) chloride;
- (c) Fe_2O_3 , diiron oxide;
- (d) BaO , barium oxide;
- (e) $\text{Mn}_3(\text{PO}_4)_2$, manganese (II) phosphate.

EXAMPLE

Name the ionic compounds



Names and Formulas of Acids

1. Acids containing anions whose names end in **-ide** are named by changing the **-ide** ending to **-ic**, adding the prefix **hydro-** to this anion name, and then following with the word acid:

Anion	Corresponding Acid
Cl^- (chloride)	HCl (hydrochloric acid)
S^{2-} (sulfide)	H_2S (hydrosulfuric acid)

2. Acids containing anions whose names end in **-ate** or **-ite** are named by changing **-ate** to **-ic** and **-ite** to **-ous** and then adding the word acid. Prefixes in the anion name are retained in the name of the acid:

Anion	Corresponding Acid
ClO_4^- (perchlorate)	HClO_4 (perchloric acid)
ClO_3^- (chlorate)	HClO_3 (chloric acid)
ClO_2^- (chlorite)	HClO_2 (chlorous acid)
ClO^- (hypochlorite)	HClO (hypochlorous acid)

Anion

____ide
(chloride, Cl^-)

add H^+
ions

Acid

hydro____ic acid
(hydrochloric acid, HCl)

____ate
(chlorate, ClO_3^-)
(perchlorate, ClO_4^-)

add H^+
ions

____ic acid
(chloric acid, HClO_3)
(perchloric acid, HClO_4)

____ite
(chlorite, ClO_2^-)
(hypochlorite, ClO^-)

add H^+
ions

____ous acid
(chlorous acid, HClO_2)
(hypochlorous acid, HClO)

EXAMPLE

1. *Name the acids*

(a) HCN,

(b) HNO₃,

(c) H₂SO₄,

(d) H₂SO₃.

EXAMPLE

*Which of the following acids are **incorrectly** named? For those that are, provide a correct name or formula.*

(a) hydrocyanic acid, HCN;

(b) nitrous acid, HNO₃;

(c) perbromic acid, HBrO₄;

(d) iodic acid, HI;

(e) selenic acid, HSeO₄.

Names and Formulas of Binary Molecular Compounds

- 1. The name of the element farther to the left in the periodic table (closest to the metals) is usually written first.*** An exception occurs when the compound contains oxygen and chlorine, bromine, or iodine (any halogen except fluorine), in which case oxygen is written last.
- 2. If both elements are in the same group, the one closer to the bottom of the table is named first.***
- 3. The name of the second element is given an -ide ending.***
- 4. Greek prefixes indicate the number of atoms of each element.***

Cl_2O dichlorine monoxide

NF_3 nitrogen trifluoride

N_2O_4 dinitrogen tetroxide

P_4S_{10} tetraphosphorus decasulfide

EXAMPLE

Name the compounds

(a) SO_2 ,

(b) PCl_5 ,

(c) Cl_2O_3 ,

(d) CS_2 ,

(e) CO ,

(f) C_3O_2 ,

(g) CBr_4 ,

(h) CF

EXAMPLE

Give the chemical formulas for

- (a) silicon tetrabromide,
- (b) disulfur dichloride,
- (c) diphosphorus hexaoxide.

2-9 Some Simple Organic Compounds

This is a self-reading section

Homeworks

Exercises:

2.6,
2.13,
2.23,
2.35,
2.49,
2.53,
2.63,
2.71,



© Pearson Education Limited 2018

You will have a quizz next week

What can you use for assistance?

- A pen
- A calculator
- A periodic table
- An A4-sized sheet of hand written notes

No other devices are allowed!